



Second-Round Interview Task Guide – Learning Data Analyst

For the second round of interviews, we kindly invite you to prepare a 15-minute presentation (3-6 slides) addressing the scenario outlined below. To respect your time, please limit your preparation to 2–3 hours. The goal of this task is to serve as a starting point for our discussion, so concise bullet points and a practical overview are all we need.

Scenario - Measuring the Development of Computational Thinking and Digital Competencies

Our institution has launched a university-wide initiative to help students from diverse academic backgrounds, e.g., Computer Science, Humanities, Arts, Social Sciences, and Business, develop computational thinking and digital competencies.

The challenge is that students enter with very different levels of digital literacy, prior experience, and learning needs. University leadership wants to understand whether the program is effective, which student groups are progressing most and least, and what interventions may be needed to support learners.

Multiple potential data sources are available, such as diagnostic assessments at program entry, learning management system activity, course assessments and project work.

Your Task

Without writing any code, design a comprehensive data pipeline architecture and analytics framework that enables the institution to measure, monitor, and improve students' development of computational thinking and digital competencies throughout their learning journey.

During the presentation explain

1. Your proposed data collection, data quality and measurement strategy.
2. The data architecture and analytics approach.
3. Key metrics and dashboards.
4. How the institution can use insights to continuously improve student outcomes.